

Lecture 7

Monotone Comparative Statics

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What Are Comparative Statics

The main way to derive testable implications from models

How equilibrium outcomes change in response to changes in parameters of the model

- How does the level of voter turnout change as party polarization changes?
- How does the level of deference to precedent change as the level of judicial independence changes?
- How does party cohesion change as the level of electoral competitiveness changes?
- How does the level of public goods provided by a government change as the size of the winning coalition changes?

Approaches to Comparative Statics

Broadly speaking, there are three approaches to deriving the comparative static implications of game theoretic models:

1. Study models with explicit solution, differentiate
2. Use the implicit function theorem to study models with implicit solutions (e.g., first-order conditions)
3. Use order-theoretic concepts to study a broader class of models

Latter two have two advantages:

1. We typically do not know the “correct” functional form, so desirable for predictions NOT to depend critically on functional forms
2. More general functional forms force the analyst to really understand the *substantive intuition* underlying the result

A Simple Example with Explicit Solutions

A sequential contest with partially interdependent valuations and quadratic cost of effort

Pl. 1 chooses a level of effort

Pl. 2 choose a level of effort

Payoffs:

$$U_1(a_1, a_2, \theta_1, \theta_2) = a_1(\theta_1 + a_2 - a_1)$$

$$U_2(a_1, a_2, \theta_1, \theta_2) = a_2(\theta_2 + a_1 - a_2)$$

Solving the Example

The optimal a_2 is given by:

$$\frac{\partial U_2}{\partial a_2} = \theta_2 + a_1 - 2a_2^* = 0 \iff a_2^* = \frac{\theta_2 + a_1}{2}$$

Now we could substitute into first player's maximization problem. Let's not yet. The first player's problem is:

$$\max_{a_1} a_1 (\theta_1 + a_2^* - a_1),$$

and the first-order condition is

$$\frac{\partial U_1}{\partial a_1} = \theta_1 + a_2^* + a_1^* \frac{\partial a_2^*}{\partial a_1} - 2a_1^* = 0$$

solve explicitly for $\frac{\partial a_2^*}{\partial a_1} = 1/2$ and substitute

$$a_1^* = \theta_1 + \frac{\theta_2}{2} \quad a_2^* = \frac{\theta_1}{2} + \frac{3\theta_2}{4}.$$

Comparative Statics in the Example

Both a_1^* and a_2^* are clearly increasing in θ_1 and θ_2

These are nice results. However, we might worry robustness and generality

In solving for the equilibrium, we did lots of canceling and substituting. When you do this, it makes the algebra easier, but it can also mask important substantive effects that you would like to know about

Some of those cancellations may only work because of particular, and simple, linear/quadratic functional form

Ideally, in order to know what is really going on in the model would like to solve something more general

$$U_1(a_1, a_2, \theta_1) = f(a_1, a_2, \theta_1) - c(a_1),$$

$$U_2(a_1, a_2, \theta_2) = g(a_1, a_2, \theta_2) - c(a_2)$$

The Example in More Generality

Make assumptions to guarantee we have an equilibrium:

$f(\cdot)$ and $g(\cdot)$ increasing and concave in a_1 and a_2 , respectively and $c(\cdot)$ increasing and convex

Player 2's first-order condition:

$$g_2(a_1, a_2^*, \theta_2) - c'(a_2^*) = 0$$

Player 1's first-order condition:

$$f_1(a_1^*, a_2^*, \theta_1) + f_2(a_1^*, a_2^*, \theta_2) \frac{\partial a_2^*}{\partial a_1} - c'(a_1^*) = 0$$

1. 1's effort directly changes production.
2. Affects the optimal choice by player 2, which then affects production
3. Costs of effort

Explicit Solutions Won't Do

In order to understand this model, we need to know how a change in effort by player 1 affects effort by player 2

would like to try to say something about how the parameters affect the optimal choices

We can't solve explicitly, so we need another method

Implicit Functions

The first-order conditions in the above example define two implicit functions
Implicitly characterize a choice, but do not provide an explicit solution in the
form of $a_1 = x$

Harder to work with, but inevitable

There are powerful tools for thinking about these functions. The most important
is the Implicit Function Theorem which allows us to calculate the derivatives of
implicitly defined variables

Building the Implicit Function Theorem

Suppose we have a function as follows:

$$G(x, \theta) = c$$

Implicitly defines an x for each θ

Let $x^*(\theta)$ be the solution of (1) as a function of θ . Then we can write:

$$G(x^*(\theta), \theta) = c$$

Want to compute $\frac{\partial x^*}{\partial \theta}$

Building the Implicit Function Theorem (cont'd)

For any pair $(\theta, x^*(\theta))$ we have $G(x^*(\theta), \theta) = c$

Therefore,

$$\frac{dG(x^*(\theta), \theta)}{d\theta} = 0$$

Using the chain rule

$$\frac{dG(x^*(\theta), \theta)}{d\theta} = \frac{\partial G(x^*(\theta), \theta)}{\partial x} \frac{\partial x^*(\theta)}{\partial \theta} + \frac{\partial G(x^*(\theta), \theta)}{\partial \theta} = 0$$

Rearranging :

$$\frac{\partial x^*(\theta)}{\partial \theta} = - \frac{\frac{\partial G}{\partial \theta}(x^*(\theta), \theta)}{\frac{\partial G}{\partial x}(x^*(\theta), \theta)}$$

We have an explicit solution for the derivative of $x^*(\theta)$.

The Implicit Function Theorem

Theorem

Let $G(x, \theta) : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuously differentiable function. Suppose that $G(x_0, \theta_0) = c$ and consider the expression $G(x, \theta) = c$. If $\frac{\partial G}{\partial x}(x_0, \theta_0) \neq 0$, then there exists a continuously differentiable function $x^*(\theta) = x^*$ defined on an interval I about the point θ_0 such that:

1. $G(x^*(\theta), \theta) = c$ for all $\theta \in I$,
2. $x^*(\theta_0) = x_0$
3. $\frac{\partial x^*(\theta_0)}{\partial \theta} = -\frac{\frac{\partial G}{\partial \theta}(x_0, \theta_0)}{\frac{\partial G}{\partial x}(x_0, \theta_0)}$

Example 1

Consider the function $x - 2\theta = 3$. Defines, for each θ , and x . We could, of course, solve explicitly, but let's not for the moment

We can write $G(x, \theta) = x - 2\theta$ and now we have an implicit function $x^*(\theta)$ with $G(x^*(\theta), \theta) = 3$

Want derivative of x^* with respect to θ . Implicit function theorem says that, if the derivative exists, it is given by:

$$\frac{\partial x^*(\theta)}{\partial \theta} = - \frac{\frac{\partial G}{\partial \theta}}{\frac{\partial G}{\partial x}}$$

Well, $G_x = 1$ and $G_\theta = -2$ So we have

$$\frac{\partial x^*(\theta)}{\partial \theta} = - \frac{-2}{1} = 2$$

Now we can confirm that this is correct by solving explicitly for x in terms of θ .

We have $x^*(\theta) = 3 + 2\theta$. So $\frac{\partial x^*}{\partial \theta} = 2$

Example 2: Implicit functions and first-order condition

Most common type of implicit function that we'll deal with is first-order condition. Consider the problem

$$\max_a f(a, \theta) - c(a)$$

with $f(\cdot)$ increasing and concave, $c(\cdot)$ increasing and convex

First-order condition is:

$$f_a(a^*, \theta) - c'(a^*) = 0$$

(where f_a is the derivative of f with respect to a)

The expression implicitly defines $a^*(\theta)$, and we have

$$G(a^*(\theta), \theta) = f_a(a^*, \theta) - c'(a^*) = 0$$

How does optimal choice a^* responds to changes in the parameter θ ?

Example 2 (cont'd)

$$G(a^*(\theta), \theta) = f_a(a^*, \theta) - c'(a^*) = 0$$

By implicit function theorem

$$\frac{\partial a^*}{\partial \theta} = -\frac{G_\theta}{G_a} = \frac{-f_{a\theta}}{f_{aa} - c''}$$

Notice our assumptions imply that $f_{aa} < 0$ and $-c'' < 0$, so the derivative has the same sign as $f_{a\theta}$

Our Old Friend the Cross-Partial

The fact that the second-order condition holds implies that the denominator is negative. Thus, the sign of the comparative static is driven by the sign of the numerator: $f_{a\theta}$

- Optimal a is increasing in θ if the cross-partial of the payoff function $f_{a\theta}$ is positive (and is decreasing if it is negative)
- $f_{a\theta} > 0$ means that as θ increases, the marginal product of an increase in a grows. Higher θ makes investment in a more profitable. When two elements of a model are related in this way, we call them complements
- Examples: education and intelligence (the smarter you are, the more you get out of time spent in school), guns and bullets (the more guns you have, the more useful additional bullets are), etc.
- a is increasing in θ if they are complements and decreasing in θ if they are substitutes

But wait...

But there is a class of problems in which we can “sign” the comparative static without actually solving the problem!

Yes, and it turns out one does not even need differentiability, or a monotonic marginal effect...

No time, but take a look at Ashworth and Bueno de Mesquita (2006) *AJPS*

Additional references for basic techniques: Milgrom and Shannon (1994); Milgrom (1994); Shannon (1995); Topkis (1998); Edlin and Shannon (1998); Athey (2002); Quah (2007); Quah and Strulovici (2009, 2012); Kukushkin (2013)